CDS528 GROUP PROJECT

SafeVendor

A trust-minimized token sale with direct 2-of-3 treasury settlement

Repository: https://github.com/aurora371 624-star/safevendor

1,000,000	2 of 3	0.0001 ETH	17
fixed SVT supply	treasury approvals	price per SVT	automated tests

Item	Submission information
Course	CDS528 Blockchain DeFi Smart Contract Development
Group	SafeVendor Group (ID: SafeVendor)
Members	Yang Qingchen, Xu Fang, Wang Zijian, Wu Fei, Chen Qikai
Network	Ethereum Sepolia, chain ID 11 155 111
Submission	18 November 2026

SafeVendor addresses a concrete token-sale custody problem: buyers must currently trust a seller or operator to hold sale proceeds and later honor withdrawals. The design removes that trust window by sending ETH directly to a fixed-owner multisig treasury in the same transaction that delivers the token. The deployed system is test-only, publicly verified, and fully documented for reproduction.

1. Problem Statement

1.1 Context and users

The primary users are small project teams or DAOs that want a simple fixed-price token sale, and buyers who want assurance that sale proceeds do not remain under one person's control. The secondary user is the three-person treasury group responsible for spending proceeds after the sale.

A conventional custody flow asks buyers to send ETH to a seller-controlled address, after which the seller promises to transfer the funds into a shared account. The buyer cannot enforce that later step. A database can record the promise, but it cannot make the subsequent transfer non-discretionary.

1.2 Problem statement

How can a fixed-price ERC-20 sale guarantee that every successful purchase settles into shared custody without allowing a seller, deployer, or single treasury member to redirect or withdraw the proceeds unilaterally?

1.3 Why a smart contract

- Settlement is atomic: the payment, treasury deposit, and token delivery either all succeed or all revert.
- The 2-of-3 threshold is enforced by code rather than by an off-chain policy.
- Token supply, price, owners, and threshold are publicly inspectable and cannot be silently changed after deployment.
- Buyers can verify the deployed bytecode and contract state on Etherscan (Etherscan, 2026).

1.4 Goals and scope

Goal	Success condition
Direct settlement	Every successful buy sends ETH to Treasury with no seller withdrawal path.
Shared control	Two distinct owners must confirm before any Treasury transfer executes.
Safe failure	Wrong payment, insufficient inventory, and failed transfers revert.
Reproducibility	A new user can install, test, deploy, and verify from the README.

1.5 Constraints and assumptions

- Sepolia only; all assets are test assets with no monetary value.
- Fixed supply of one million SVT; there is no post-deployment mint.
- Three fixed treasury owners and a threshold of two.
- No proxy, upgrade path, pause switch, or owner replacement.
- The sale price does not follow a market oracle; it is a teaching mechanism.

2. Solution Design

2.1 Architecture and roles

SafeVendor separates three responsibilities. The token contract owns only the fixed supply. The vendor owns the sale rule and inventory. The treasury owns shared custody and transaction approval. No contract can upgrade or replace another contract after deployment.

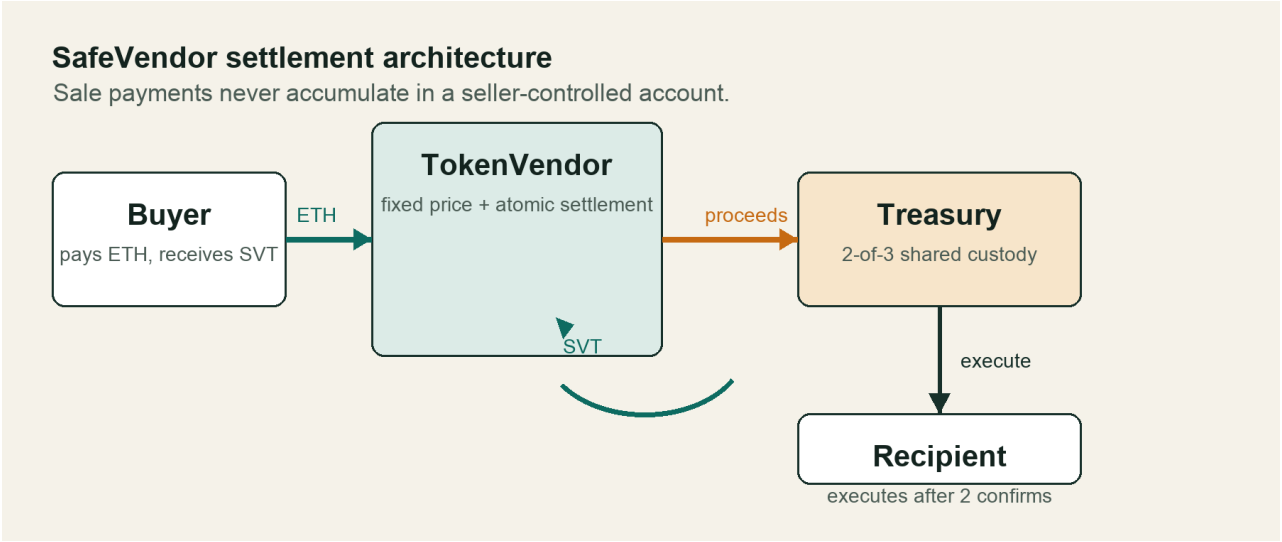


Figure 1. Sale payments settle directly in the multisig treasury.

Actor	On-chain capability	Trust boundary
Buyer	Calls buyTokens with ETH and a minimum SVT output.	Does not need to trust a seller address with custody.
Treasury owner	Submits, confirms, and executes proposals when threshold is met.	One owner is insufficient; two can collude.
Seller	Supplies token inventory; has no vendor withdrawal function.	Cannot retrieve sale ETH from the vendor.
Frontend	Reads state and prepares wallet transactions.	Cannot bypass contract access control.

2.2 Mechanism and token rationale

ERC-20 is appropriate because the sale needs a fungible unit that any wallet can hold and transfer (Ethereum Foundation, 2026). SafeVendorToken uses OpenZeppelin ERC20 and mints the entire one-million-token supply once. It deliberately omits owner-only mint and burn functions, preventing post-sale supply dilution (OpenZeppelin, 2026).

The vendor stores the token address, treasury address, and price as immutable values. A fixed price is feasible for this teaching scope and avoids oracle complexity. The design's distinctive property is not a new token standard. It is the settlement combination: a familiar vendor pattern whose proceeds flow into a multisig that no single operator controls.

2.3 Key process flows

Purchase flow	Treasury flow
1. Buyer sends ETH and minTokensOut.	1. Owner submits a recipient, value, and calldata.
2. Vendor quotes the fixed-price token amount.	2. Submission records the first owner confirmation.
3. Vendor checks inventory and slippage.	3. A different owner confirms the proposal.
4. Vendor sends net ETH to Treasury.	4. Execution requires confirmation count >= 2.
5. Vendor transfers SVT and emits an event.	5. Executed is set before the external call.

2.4 On-chain and off-chain partition

Trust-critical state and value transfer stay on-chain: balances, ownership, confirmations, execution state, and ETH movement. The optional Vite frontend stays off-chain because it only reads chain state and prepares signed transactions. There is no server-side custodian or backend signing key.

3. Implementation

3.1 Core mechanisms

Mechanism	Implementation	Why it matters
Fixed supply	One constructor mint; immutable maxSupply.	No post-deployment dilution.
Fixed price	Math.mulDiv converts ETH amount to token units; minimum output protects the buyer.	Deterministic sale terms.
Direct settlement	Checked native call forwards net ETH to Treasury before token delivery.	No seller custody window.
2-of-3 execution	Per-owner confirmation mapping and immutable threshold.	One owner cannot move funds.
Replay protection	Executed state is set before the external call.	A transaction cannot execute twice.
Reentrancy protection	CEI plus OpenZeppelin ReentrancyGuard.	External calls cannot re-enter value paths.

3.2 Events and access control

Token and vendor behavior uses standard ERC-20 Transfer events and a custom TokensPurchased event with gross payment, net settlement, refund, and token amount. Treasury emits Deposit, TransactionSubmitted, TransactionConfirmed, and TransactionExecuted. The event set is sufficient to reconstruct each purchase and approval sequence from logs.

Treasury submission, confirmation, and execution are guarded by onlyOwner. The owner mapping and fixed owner array are initialized once. The vendor has no privileged withdrawal function, so even the deployer cannot retrieve its sale ETH.

3.3 Gas-awareness and code quality

- Immutable token, treasury, price, threshold, and max supply avoid storage reads.
- The only loop is bounded owner initialization at deployment time.

- Small custom errors replace long revert strings.
- The optimizer is enabled with 200 runs.
- Events expose value-bearing state changes without requiring historical storage.
- Code is split by responsibility and uses OpenZeppelin components rather than reimplementing token, transfer, and reentrancy primitives.

3.4 Frontend integration

The dApp displays live Sepolia state, quotes a purchase, collects wallet signatures, submits treasury proposals, and exposes owner confirm and execute actions. It reads deployment metadata from a generated JSON file, so the interface cannot silently point at a different contract set. The interface uses Vite and ethers.js (Ethers, 2026; Vite, 2026).

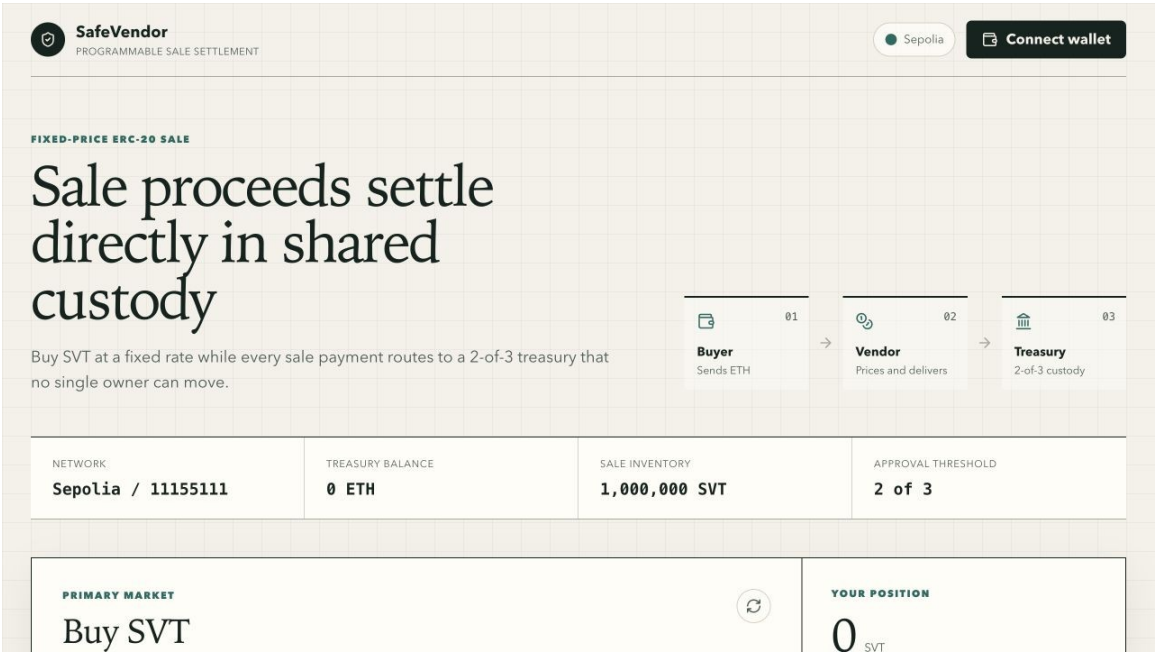


Figure 2. Operational dApp displaying live Sepolia sale and treasury state.

4. Testing and Security

4.1 Verification results

17	100%	100%	2
tests passing	production line coverage	function coverage	Slither INFO findings

The test suite exercises successful behavior and failure paths, including duplicate confirmation, unauthorized callers, replay, slippage, insufficient inventory, invalid configuration, rounding refunds, and reentrancy. Production contracts have 100 percent statement, function, and line coverage. Aggregate branch coverage is 79.03 percent.

4.2 Static analysis

Slither (Trail of Bits, 2026) analyzed 18 compiled contracts with 100 detectors after excluding OpenZeppelin dependencies and the test-only receiver. It reported no High or Medium severity findings. The two project-level findings are informational low-level call findings.

The scan was executed from the repository root with the following command. The relevant output is reproduced below the command.

```
PYTHONPATH=work/slither-packages python -m slither . \
  --filter-paths "contracts/test" \
  --exclude-dependencies

INFO:Detectors:
Low level call in MultiSigTreasury.executeTransaction(uint256)
Low level call in TokenVendor.buyTokens(uint256)
INFO:Slither:. analyzed (18 contracts with 100 detectors), 2 result(s) found
```

Finding	Location	Decision
Low-level call	MultiSigTreasury.executeTransaction	Required for arbitrary approved calldata and contract wallets; success is checked.
Low-level call	TokenVendor.buyTokens	Required for ETH settlement and refunds; both return flags are checked.

4.3 Top threats and mitigations

Threat	Attack path	Mitigation and residual risk
Single-owner exfiltration	One compromised owner submits and executes a transfer.	2-of-3 threshold blocks one owner; two colluding owners remain a residual risk.
Reentrancy	Recipient re-enters during ETH delivery.	CEI plus ReentrancyGuard; the reentry test confirms rejection.
Replay	Execute the same approved transfer twice.	Executed is checked and set before the call; replay reverts.
Duplicate approval	One owner confirms repeatedly to reach threshold.	A per-owner, per-transaction confirmation mapping blocks duplicates.
Unauthorized action	Non-owner submits, confirms, or executes.	OnlyOwner on all treasury state changes.
Payment ambiguity	Buyer overpays or price rounding differs.	Minimum output and explicit remainder refund; failed sends revert.

4.4 Economic and compliance risks

- The fixed price is not market-derived and can diverge from any external value.
- Three EOAs create key-management risk; the design assumes independent custody.
- The absence of an upgrade path prevents patching; migration requires redeployment.
- There is no owner-removal mechanism; lost keys reduce availability.
- The project must remain on Sepolia and must not be presented as an investment.

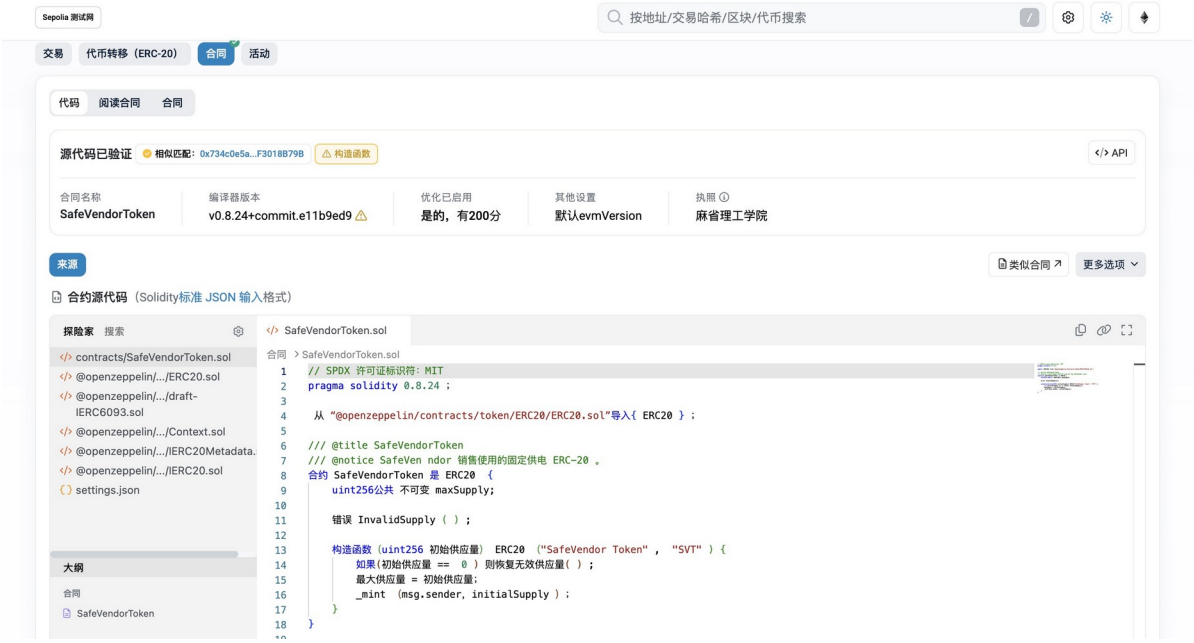


Figure 3. Etherscan reports verified source, ABI, and compiler settings for SafeVendorToken.

5. Deployment and DevOps

5.1 Live Sepolia deployment

The contracts are deployed on Sepolia with Solidity 0.8.24, optimizer enabled, and 200 runs. Etherscan matches the deployed bytecode to verified source and ABI; the Treasury page uses Etherscan's similar-bytecode source match.

Contract	Verified Sepolia address
SafeVendorToken	0xb2caA86b7b758 800C0 992 371 621 3719C456F D46E
MultiSigTreasury	0x427 038b1e08c3C582C6 096 233 285bd7b26fab4 c9
TokenVendor	0x1BE452F65B35E3aa8ecD41BbCD7d7C9 324 70 5270

5.2 Reproducible deployment

The Hardhat deployment scripts (Nomic Foundation, 2026) validate chain ID 11 155 111, deploy Token, Treasury, and Vendor in dependency order, transfers the full inventory, and writes addresses, transaction hashes, and constructor data to deployments/sepolia.json. The frontend imports the generated deployment file so UI state and on-chain contracts stay aligned.

Artifact	Purpose
pnpm compile	Compiles Solidity 0.8.24 with optimized settings.
pnpm test	Runs 17 unit and failure-path tests.
pnpm coverage	Writes Istanbul coverage reports.
pnpm deploy	Deploys and records addresses and receipts.
pnpm verify	Verifies constructors through the Etherscan API.
pnpm frontend:build	Creates the production Vite bundle.

5.3 Operational readiness

The contracts deliberately omit pause and upgrade controls. This choice removes a privileged emergency key and prevents an administrator from freezing the sale or replacing the treasury. The trade-off is that a logic defect cannot be patched in place; the team would deploy a new immutable contract set and inform users.

Operationally, two of three owner keys must remain available. Only one may be offline without blocking execution. Two offline or lost keys halt treasury spending, while two compromised keys can collude. These conditions are stated explicitly rather than hidden behind an admin recovery function.

5.4 Live deployment evidence

Event	Observed result
Owner 1	0x758 090f1c15a76fFf31C5852e71f68ABCF24FA61
Owner 2	0xbaA884A7B6cAe7d8CEC920a5Ce92e47B6B231 031

Event	Observed result
Owner 3	0x8fcFd31be53b4b2F8C9394d49f92dE56Da2b75c2
Threshold	2 of 3 confirmations required.
Vendor inventory	1,000,000 SVT received at deployment.
Source verification	All three contracts show verified source and ABI on Etherscan; the Treasury is a similar-bytecode match.

6. Documentation, Process, and Evaluation

6.1 Repository and documentation

The repository includes source, tests, deployment scripts, a reproducible README, security evidence, a requirements matrix, a demo script, and the optional frontend. The readme identifies exact prerequisites, network, commands, risk controls, and the operational rationale for immutability.

The public GitHub URL is on the cover and at the top of the repository README. The cover identifies the group name, group ID, and all five members.

6.2 Rubric coverage

Criterion	Evidence	Risk
R1 Problem Statement	Concrete users, custody pain point, why on-chain, scope and assumptions.	Low
R2 Solution Design	Three-contract architecture, actor table, rationale, purchase and approval flows.	Low
R3 Implementation	Deployed contracts, events, access control, immutable values, frontend.	Low
R4 Testing and Security	17 tests, full production line coverage, Slither, threat table.	Low
R5 Deployment and DevOps	Three verified Sepolia addresses, reproducible scripts, immutability rationale.	Low
R6 Documentation	README, report, citations, license, AI disclosure.	Low
R7 Presentation	Editable slide deck, 12-minute timing, five named speaker roles, Q&A preparation.	Low

6.3 Team process

- Yang Qingchen, Group Lead: scope, architecture review, submission control, and slides 1-2.
- Xu Fang, Smart Contract Lead: contract design, purchase flow, code review, and slides 3-4.
- Wang Zijian, Testing and Security Analyst: treasury flow, failure paths, Slither evidence, and slides 5-6.
- Wu Fei, Deployment and DevOps Engineer: verification, deployment, Etherscan evidence, and slides 7-8.
- Chen Qikai, Frontend and Documentation Engineer: dApp workflow, deployed state, conclusion, and slides 9-10.
- Each member presents a two-slide segment and answers at least one question.

6.4 Limitations and future work

The current project intentionally favors a small auditable scope. Future work could add an explicit inventory recovery decision, owner rotation with timelocks, proposal cancellation, richer event indexing, fuzz invariants, and a formal threat model for market pricing. Those additions would introduce new trust surfaces and should not be added without matching tests and governance rationale.

6.5 Academic integrity and licensing

AI-assisted tools were used for implementation drafts, test ideation, report editing, and presentation drafting. The group reviewed the generated material, ran the tests, inspected the deployment evidence, and checked the licenses before submission. AI output was not treated as evidence of technical correctness.

The project-authored source and documentation use the MIT License. Third-party software retains its upstream license. The table records the principal software used in the delivered system.

Component	Delivered version	License
OpenZeppelin Contracts	5.4.0	MIT
Hardhat	2.26.3	MIT
ethers.js	6.15.0	MIT
Vite	7.1.7	MIT
Lucide	0.468.0	ISC
Slither	0.11.3	AGPL-2.0

7. Conclusion

SafeVendor meets the assignment's central engineering goal. It presents a concrete DeFi custody problem, implements a complete Solidity system, tests failure paths, deploys and verifies on Sepolia, provides a usable frontend, and documents the operational trade-offs. The architecture is deliberately small: three contracts with clear ownership boundaries and no hidden administrator.

The strongest evidence is the live Sepolia state: the group owner is fixed at deployment, the treasury requires a second independent owner, the vendor holds the complete token inventory, and Etherscan shows matching source and ABI for all three contracts.

The report and presentation include the group identity and a complete speaker map. The remaining work is the timed rehearsal and the two Moodle submissions.

References

Ethereum Foundation. (2026). Ethereum developer tutorials. https://ethereum.org/developers/tutorials/

Ethereum Foundation. (2026). ERC-20 token standard. https://ethereum.org/developers/docs/standards/tokens/erc-20/

Etherscan. (2026). Ethereum Sepolia block explorer. https://sepolia.etherscan.io/

Ethers. (2026). ethers.js documentation. https://docs.ethers.org/

Nomic Foundation. (2026). Hardhat documentation. https://hardhat.org/docs

OpenZeppelin. (2026). OpenZeppelin Contracts 5.x. https://docs.openzeppelin.com/contracts/5.x/

Solidity Team. (2024). Solidity 0.8.24 documentation. https://docs.soliditylang.org/en/v0.8.24/

Trail of Bits. (2026). Slither static analysis. https://github.com/crytic/slither

Vite. (2026). Vite documentation. https://vite.dev/

Appendix A. Deployment Addresses

Contract	Verified address	Deployment transaction
SafeVendorToken	0xb2caA86b7b758 800C0 992 37 16 213 719C456FD46E	0x473a9b7441b61b8da19af84b16d6bd1ac5aed13f08 716d331ddb56 796b2b53dc
MultiSigTreasury	0x427 038b1e08c3C582C6 096 23 3285bd7b26fab4c9	0xa0e81 226b1b899 582a568f9b77f5995a24d6d41 21 3df54 405b6574c9c8 470 774
TokenVendor	0x1BE452F65B35E3aa8ecD41B bCD7d7C9 324 705 270	0x1d02dd7bb4c18c7f5afb08e50 533 729e77f8c67f89 7905e30e1ed24c72 620 818